

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: Dirac- δ structure, Heaviside-type cancellation, and learned smooth correction

HYBRID GREEN KERNEL

$$G_{\theta} = \text{Dirac-}\delta \text{ structure} + \text{Heaviside-type cancellation} + \text{learned smooth correction.}$$

DIRAC- Δ STRUCTURE

Provide the source singularity.

HEAVISIDE-TYPE CANCELLATION

Cancel the step-like contribution induced by the jump construction.

LEARNED SMOOTH CORRECTION

Learn the remaining smooth residual.

Built-in singular and cancellation structure reduces what the neural network must learn.

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$$G_{\theta}(t, \eta) = \underbrace{E(t, \eta)M(t)R_{\theta}(t, \eta)}_{\text{learned smooth correction}} + \underbrace{B(t) \left(J_0(t, \eta) - \frac{1}{2}E(t, \eta) \right)}_{\text{Heaviside-type cancellation}} + \underbrace{A(t)G_0(t, \eta)}_{\text{Dirac-}\delta \text{ jump}}.$$

DIRAC- Δ STRUCTURE

$$G_0(t, \eta) = \begin{cases} t(1 - \eta), & t < \eta, \\ \eta(1 - t), & t \geq \eta, \end{cases} \quad \partial_t^2 G_0 = -\delta(t - \eta).$$

HEAVISIDE-TYPE CANCELLATION

$$\partial_t J_0(t, \eta) = G_0(t, \eta), \quad S = J_0 - \frac{1}{2}E, \quad \partial_t^2 S = \partial_t G_0.$$

COEFFICIENT FACTORS

$$A(t) = \frac{1}{a_{\text{unit}}(t)}, \quad B(t) = \frac{a'_{\text{unit}}(t) + b_{\parallel, \text{unit}}(t)}{a_{\text{unit}}(t)^2}.$$

The analytic terms handle the distributional Green-function behavior before learning.

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LEARNED SMOOTH CORRECTION

$$E(t, \eta)M(t)R_{\theta}(t, \eta), \quad E(t, \eta) = t\eta(1 - \eta), \quad M(t) = 1 - t.$$

GreenNet learns the smooth correction R_{θ} ; the singular and cancellation structure is built into the kernel form.

The neural correction focuses on the remaining smooth part of the kernel.